FINM3013 Introduction to Financial Derivatives

Chapter 5 Exercises

1. The current price of silver is $30 per ounce. The storage costs are $0.48 per ounce per year payable quarterly in advance. Assuming that interest rates are 10% per annum for all maturities, calculate the futures price of silver for delivery in nine months.

2. In early 2012, the spot exchange rate between the Swiss Franc and U.S. dollar was 1.0404 ($ per franc). Interest rates in the U.S. and Switzerland were 0.25% and 0% per annum, respectively, with continuous compounding. The three-month forward exchange rate was1.0300 ($ per franc). What arbitrage strategy was possible? How does your answer change if the exchange rate is 1.0500 ($ per franc).

3. A stock is expected to pay a dividend of $1 per share in two months and in five months. The stock price is $50, and the risk-free rate of interest is 8% per annum with continuous compounding for all maturities. An investor has just taken a short position in a six-month forward contract on the stock.
   a) What are the forward price and the initial value of the forward contract?
   b) Three months later, the price of the stock is $48 and the risk-free rate of interest is still 8% per annum. What are the forward price and the value of the short position in the forward contract?

4. A trader owns a commodity as part of a long-term investment portfolio. The trader can buy the commodity for $950 per ounce and sell it for $949 per ounce. The trader can borrow funds at 6% per year and invest funds at 5.5% per year. (Both interest rates are expressed with annual compounding.) For what range of one-year forward prices does the trader have no arbitrage opportunities? Assume there is no bid–offer spread for forward prices.